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Introduction to LangChain

Instructor

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Outline

- What is LangChain
- The LangChain Core Framework
- LLM Input / Output with LangChain
- Retrieval with LangChain
- Tools and Agents with LangChain
- Building Chains with LangChain
- Other features

What is LangChain

- An open framework & ecosystem to develop applications powered by LLMs using a suite of tools

What is LangChain

- Its core framework consists of libraries to connect with LLMs using prompts and format outputs using output parsers.
- Multiple steps can be chained and complex conversational applications, RAG systems, and agents can be built.

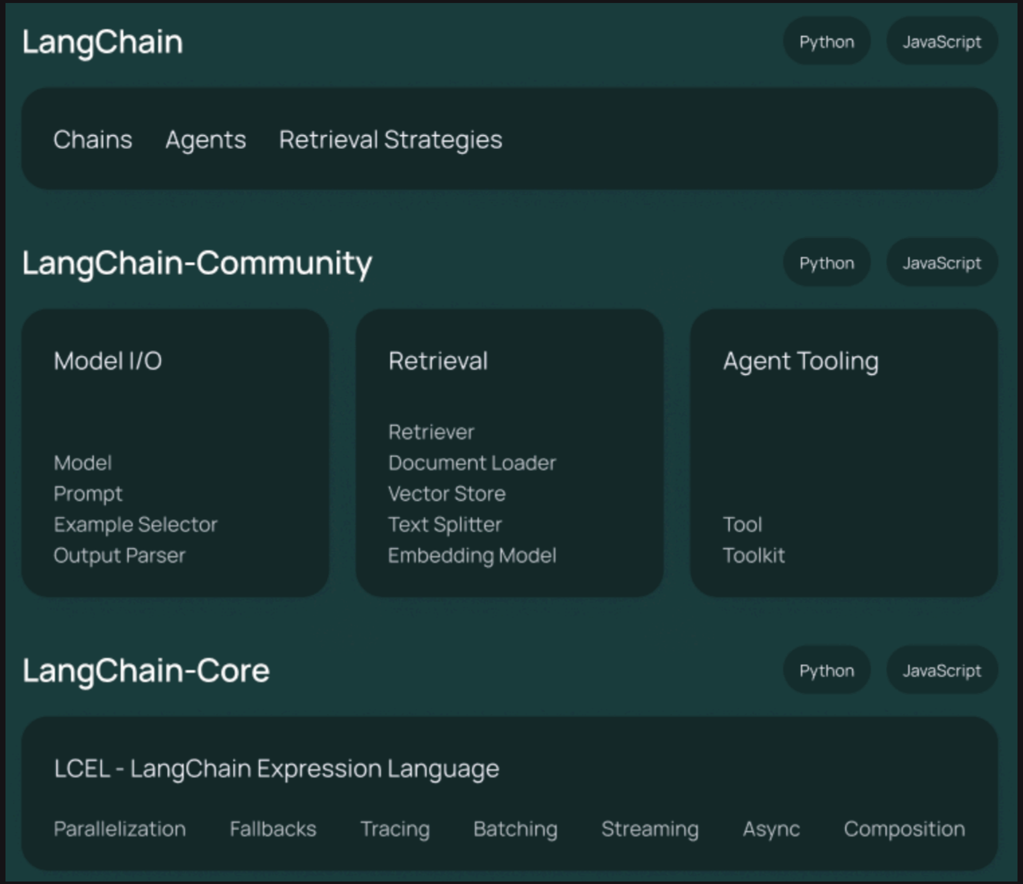
What is LangChain

- LangChain features LangServe and LangSmith for deploying, evaluating, and monitoring LLMs.
- LangChain includes LangGraph, which allows for combining multiple chains to build complex agents and more.

LangChain Core Framework

It is a framework that gives developers and data scientists a suite of components to build LLM applications easily.

- As the name suggests, one can easily chain multiple operations together to build simple as well as complex conversational LLM applications.
- Compatible with almost all popular large language models.



LLM Input/Output with LangChain

langchain-openai

langchain-google-genai

langchain-aws

langchain-google-vertexai

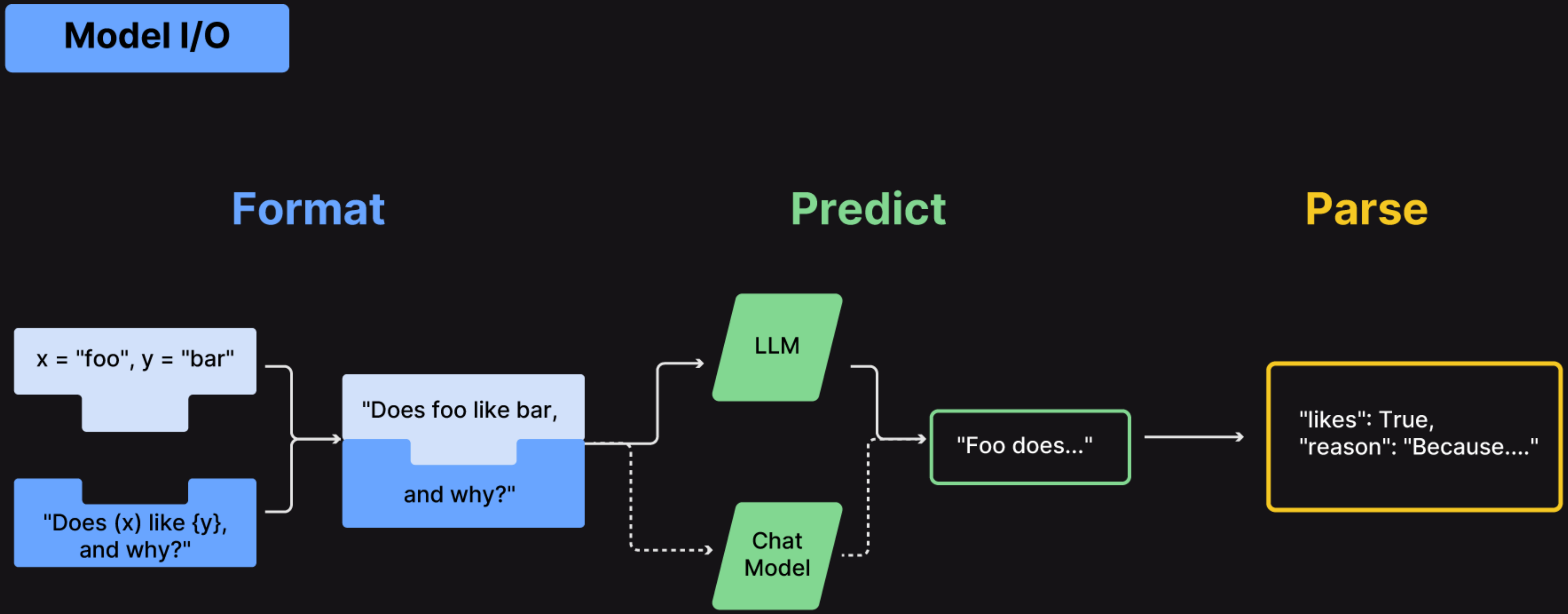
langchain-anthropic

LangChain has the necessary libraries and functions to connect with all major closed LLMs like ChatGPT, Gemini, Claude and popular models from cloud providers.

langchain-community

LangChain has a community library which is a one-stop shop to connect with all open LLMs including popular models hosted on HuggingFace, Vector DBs and more

LLM Input/Output with LangChain



LLM Input/Output with LangChain

- LangChain provides interfaces to talk to LLMs using its popular ChatModel construct and predict responses based on input prompts
- Constructs like PromptTemplate and ChatPromptTemplate allow for dynamic, data-driven prompt formatting and creation at runtime.

LLM Input/Output with LangChain

- Allows access to output parsers to parse LLM responses into formats like JSON, CSV, and more
- Enables us to do iterative prompt engineering easily

Why Retrieval with LangChain

- Many enterprise applications use LLMs to access custom enterprise data for answering questions beyond their original training data.

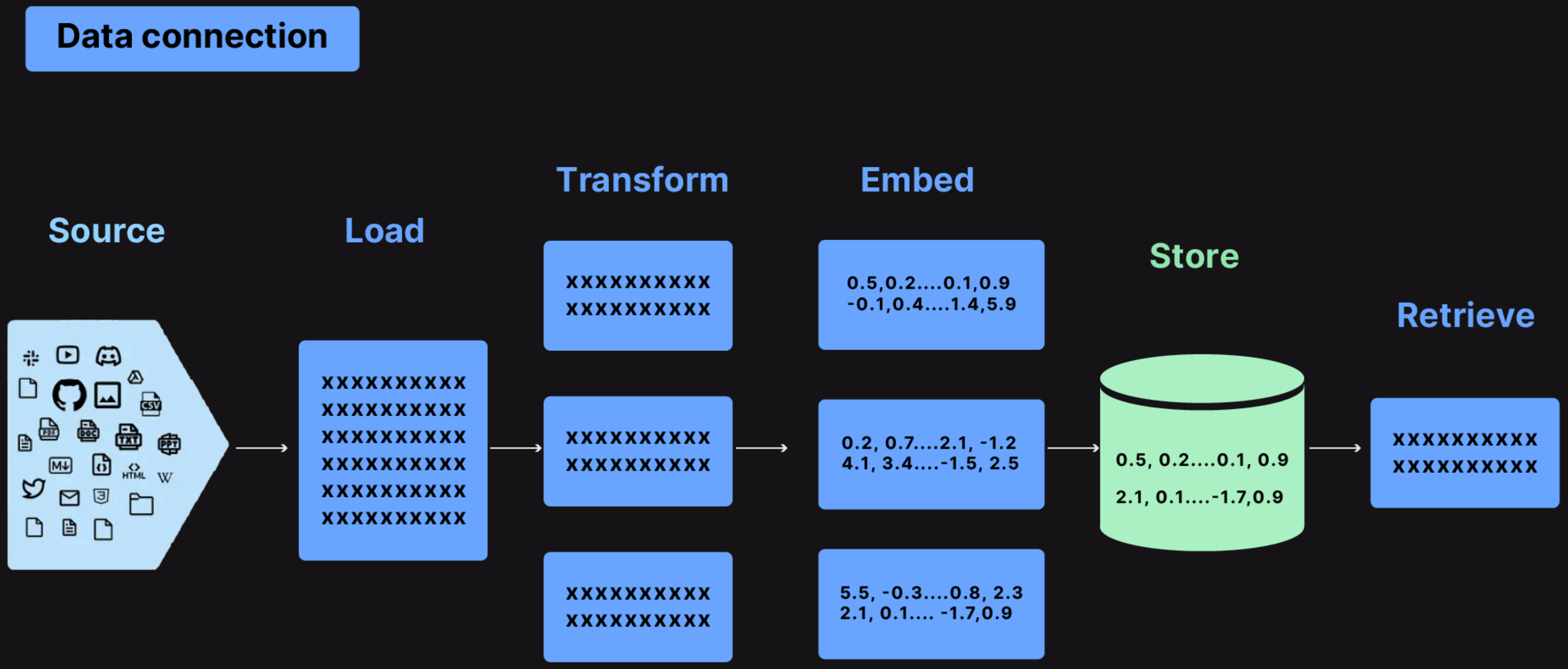
Retrieval Augmented Generation (RAG) System

A system which integrates custom data and augments an LLM's knowledge-base to generate contextual responses

Why Retrieval with LangChain

- LangChain helps in building both the retrieval and the response generation pipeline

Retrieval with LangChain



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Retrieval with LangChain

- LangChain provides over 100 different document loaders:
  - as well as integrations with other major providers like Unstructured to load data from all types of documents (HTML, PDF, code)
- Provides several transformation algorithms:
  - to split, chunk and transform larger documents into more manageable chunks for retrieval

Retrieval with LangChain

- Offers integrations with more than 25 embedding model providers:
  - ranging from open-source solutions like HuggingFace to proprietary ones like OpenAI
  - enabling the embedding of document chunks into vector embeddings.

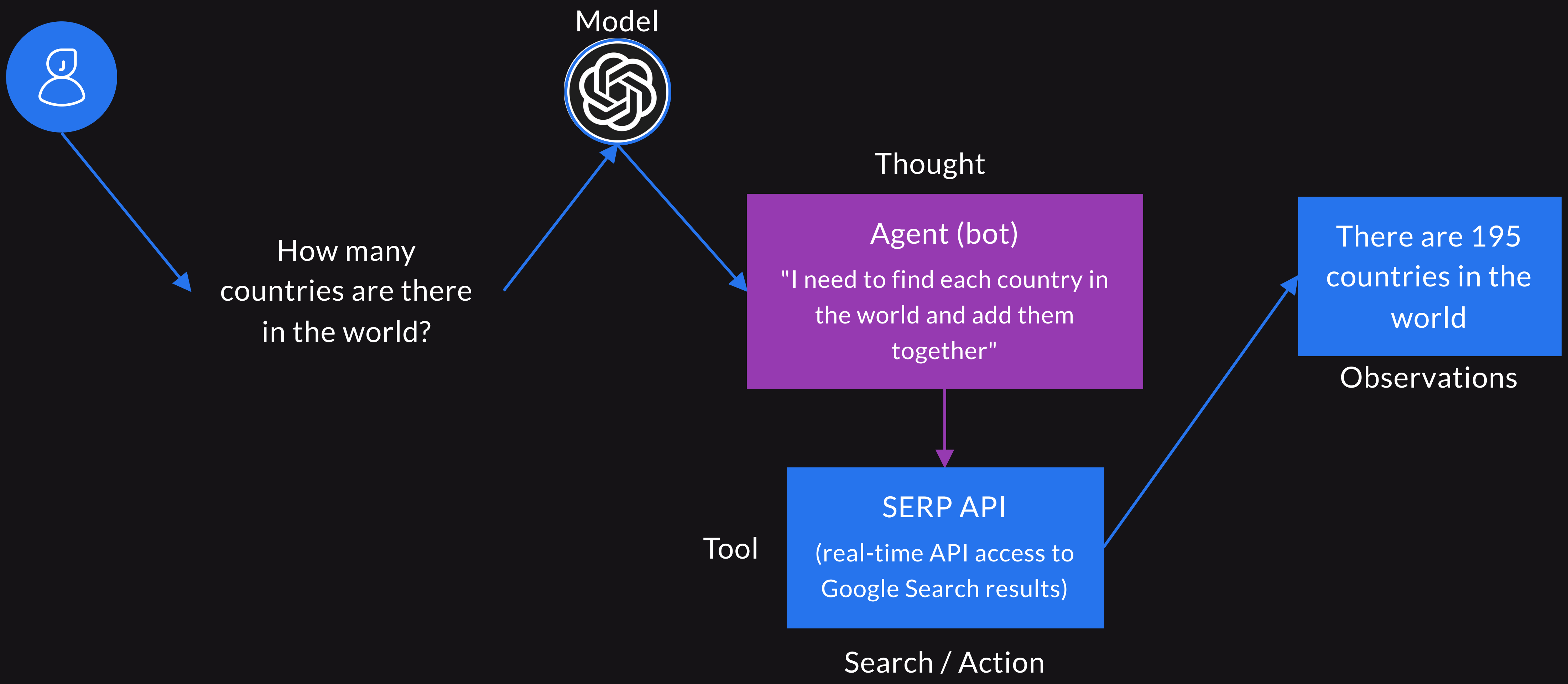
HuggingFace

OpenAI

Retrieval with LangChain

- Integrates with over 50 vector database providers:
  - to store document chunks and embeddings.
- Supports many different retrieval and search algorithms:
  - to retrieve relevant document chunks based on user queries

Tools and Agents with LangChain



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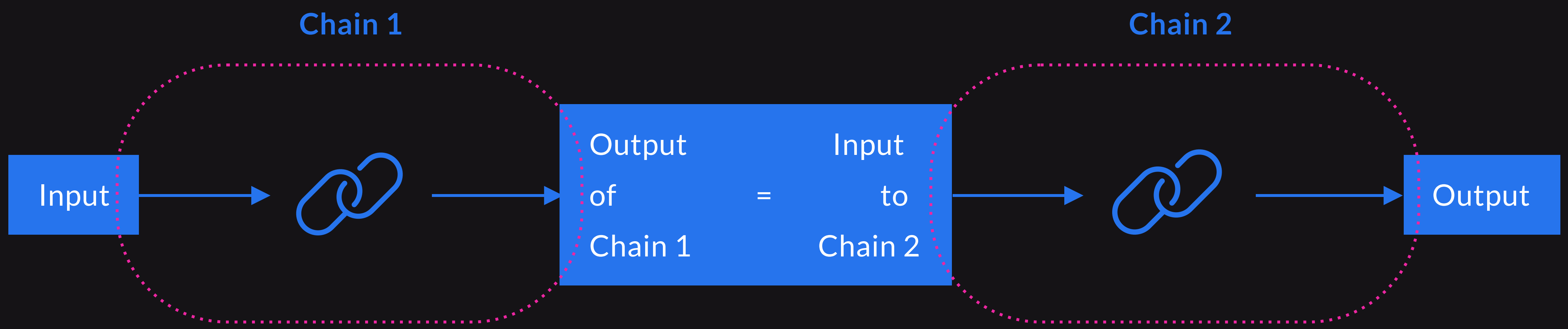
Tools and Agents with LangChain

- LangChain enables to build tools that give an interface for LLMs and other components to interact with other systems
  - Example tools include Wikipedia, calculator, search engines, and more
- Allows for the creation of agents that use LLMs to determine actions, which are executed using various tools.
  - An executor serves as the runtime environment that facilitates the execution of these agents.

Building Chains with LangChain



Building Chains with LangChain

- LangChain allows the use of chains, which are sequences of calls or execution steps.
  - These calls can be made to an LLM, a tool, or a data preprocessing step.

Building Chains with LangChain

LangChain supports two major types of chains:

- ## Legacy Chains
  - These are derived classes constructed by subclassing a legacy Chain class
  - These chains do not use LCEL under the hood but are the standalone classes
- ## LCEL Chains
  - It uses the newer LangChain Expression Language (LCEL) to build a chain of steps by using the overloaded vertical bar or pipe '|' operator

Other features

- Enables to connect to almost any third-party LLM, Vector DB, Embedding models, and Data loaders
- Supports storing LLM conversation history using memory constructs, either in-memory, on disk, or in a data store.
  - It integrates with all popular data stores for effective memory management.
- Has capabilities for callbacks which are useful for logging, monitoring, streaming, and other tasks

Thank You